

2026 Evil Swerve Drive Todo

File	Todo	Priority	Priority Numb	Status
TurretLauncher	Add speed control for launcher. One PID will create output value for both motors. Add motors, Add sparkmax encoders, create average velocity, add simple motor feedforward and PID. (see swervemodule drive controller.)	High	1	Partial
IntakeSystem	determine up and down angles on the robot -- suggest starting with 90 is up and 0 is down...	Low	4	Partial
TurretLauncher	Convert to degrees. Think that the default is rotations. Also take into account gear ratio...	High	1	Open
IntakeSystem	scale arm position.. default is rotations.... so $DEG = ROT * 360 / GEAR_RATIO$	High	2	Open
TrajectorySystem	Make this modulo +/- PI... There is a WPILIB function for this... MathUtil.angleModulus	High	2	Open
TurretLauncher	Remove turret motor config. Config burned into motor from Rev Hardware Client..	High	2	Open
TurretLauncher	Need a way to set turret positon demand from joystick for testing.. Increment / Decrement value. Maybe leave this as a manual mode.	High	2	Open
TurretLauncher	"readSensors" is never called. Also object should be defined in init, not during each execution. Read both launcher encoders and average values. It is okay to leave launch encoder velocity in RPM.	High	2	Open
TurretLauncher	More definition of turret position control object to init from excute	High	2	Open
AutoSystem	Add items to NT tables for debugging and watching autos. Suggest Step, Timeout, Parm1, Parm2, Parm3, Function. Add public getters for these.	Medium	3	Open
AutoSystem	Publish list of read autos to Network Tables. Use chooser to select which auto to run.	Medium	3	Open
IntakeSystem	Remove Motor config. Maybe just comment out. Motor config set via Rev Hardware Client.	Medium	3	Open
SwerveTeleop	Add a driver button to execute a trajectory while button is held done. When first pressed (edge triggered) set init variable in call to follow trajectory.	Medium	3	Open
SwerveVision	will have to set physical camera offset position once it is known. There is a spreadsheet in programming slack channel.	Medium	3	Open

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TrajectorySystem	add network table entries for ontarget, done, havesample, timelengthoftrajectory (rename if you don't like name)	Medium	3	Open
TrajectorySystem	TODO: add getters for new things written to network tables. ontarget, havesample,done.	Medium	3	Open
TurretLauncher	Set physical offset of robot to turret	Medium	3	Open
TurretLauncher	Set "dump to our side" field positions Red and Blue (also maybe left and right)	Medium	3	Open
TurretLauncher	Set on / off command for launcher.	Medium	3	Open
TurretLauncher	Add limit switch values to network tables.	Medium	3	Open
TurretLauncher	Calculate turret on target and add value to network tables. (Should this value prevent launcher from shooting ????)	Medium	3	Open
AgitatorSystem	Remove unused imports	Low	4	Open
Climb	Do the TODO... We don't know what climb is yet.	Low	4	Open
FeederSystem	Remove unused imports.	Low	4	Open
FeederSystem	Might need to do speed control here...	Low	4	Open
IntakeSystem	Say which motor is arm and which is intake.... Is 1 intake, and 2 arm ??	Low	4	Open
IntakeSystem	config values need to be changed/tuned	Low	4	Open
IntakeSystem	This appears to be redundant.	Low	4	Open
IntakeSystem	Is this intake or arm motor ????	Low	4	Open
IntakeSystem	what is this for? Maybe check the actual angle -- encoderRot -- instead.	Low	4	Open
SwerveDrive	Remove unused imports	Low	4	Open
SwerveModule	Remove unused imports	Low	4	Open
SwerveOdometry	put some comments here. For example, this is the initial robot position (pose)	Low	4	Open
SwerveTeleop	Remove old commented out things that will never be used.	Low	4	Open
SwerveTeleop	Add driver button box ???	Low	4	Open
SwerveTeleop	Add aux button box ???	Low	4	Open
TrajectorySystem	Remove unused imports.	Low	4	Open
TurretLauncher	Remove unused imports	Low	4	Open
AgitatorSystem	Add JavaDoc	Very Low	5	Open
AutoRoutine	Add JavaDoc	Very Low	5	Open
AutoStep	Suggest adding the following Cmd -- ExecuteCmd (command name is in Function string variable)	Very Low	5	Open
AutoStep	Suggest adding the following Cmd -- ExecuteCheckFor (command name is in function string variable)	Very Low	5	Open
AutoStep	Suggest adding the following Cmd -- AutoDone -- This will do nothing forever....similar to wait but with a forever timeout...	Very Low	5	Open
AutoStep	Add JavaDoc	Very Low	5	Open

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AutoSystem	Add doc -- call this just prior to running ANY auto... Robot.java has a place for this. It isn't the overall just once init.	Very Low	5	Open
AutoSystem	Does not compile. Replace with generic Exception instead of FileNotFoundException.	High	1	Complete
SwerveModule	The NEO Vortex is NOT use spark max. I think it uses a Spark Flex controller. These are different. Need to read and convert.	High	1	Complete
AutoRoutine	set class variable with value of autoDescription parameter..	High	2	Complete
AutoRoutine	set class variable with value of autoDescription parameter.. (2 nd constructor)	High	2	Complete
AutoSystem	Add code to the init function to get all available autos, then call the function to read them into the auto cache...	High	2	Complete
AutoSystem	Add private static HashMap to store all trajectories <String,Trajectory<SwerveSample>	High	2	Complete
AutoSystem	Add your newRoutine to the arrayList routines...	High	2	Complete
AutoSystem	Add your newRoutine to the class variable to be created containing the HashMap	High	2	Complete
AutoSystem	If you store the routines in the class variable containing the HashMap, then this routine doesn't need to return anything (void)	High	2	Complete
AutoSystem	Add a function to return an AutoRoutine from the hashmap with a calling String parameter of the name (file name without extension)	High	2	Complete
AutoSystem	Add call to TrajectorySystem.FollowTrajectory	High	2	Complete
IntakeSystem	Use built in spark max encoder. We are no longer getting a through bore encoder..	High	2	Complete
IntakeSystem	Encoder definition isnt correct.. Now use spark max built in encoder .GetEncoder(); see above.	High	2	Complete
IntakeSystem	limit switch is new wired to roboRIO... so this won't work. Use digital input.. Need to define object (class level), create object(in init function), read object (here).	High	2	Complete
TrajectorySystem	add calculate of "complete" and return "complete" to user. complete = elapsedtime >= trajectory time length AND onTarget.	High	2	Complete
TurretLauncher	get the values of the goal position based on alliance. I think there is a spreadsheet in slack. For now just use RED target.	High	2	Complete
TurretLauncher	This will not be throughbore encoder any longer. Use built in SparkMax encoder....	High	2	Complete
TurretLauncher	Use spark max built in encoder instead. Build didn't mount through bore encoder....	High	2	Complete
TurretLauncher	getPosition can be used above....	High	2	Complete
AgitatorSystem	Set motor CAN Id	Medium	3	Complete
AutoFunctions	Add new java file for static Auto processing functions.	Medium	3	Complete

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AutoFunctions	Add AutoFunctions.java file for static functions. Add an boolean AutoWait() function. It should set drive demand to zero and return false..	Medium	3	Complete
AutoSystem	add locNTSend trigger update.	Medium	3	Complete
AutoSystem	Add call to AutoFunctions.AutoWait().	Medium	3	Complete
FeederSystem	Set feeder motor CAN ID	Medium	3	Complete
TurretLauncher	TODO: Set CAN ID	Medium	3	Complete
AutoRoutine	Remove unused imports	Low	4	Complete

Notes
Partially done. Sensors not read. Both motor velocity not averaged. Encoder object not defined in init. No need to negate. Negate is done in motor config...
This is sort of done....

[illegible]

Needed to add this to the individual case....
